Morphomolecular Classification Update on Hepatocellular Adenoma, Hepatocellular Carcinoma, and Intrahepatic Cholangiocarcinoma

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Abbreviations: b-HCA = β-catenin–activated HCA, CCA = cholangiocarcinoma, HBP = hepatobiliary phase, HCA = hepatocellular adenoma, HCC = hepatocellular carcinoma, HHCA = *HNF1A*–inactivated HCA, iCCA = intrahepatic CCA, IHCA = inflammatory HCA, MF = mass forming, MTM = macrotrabecular massive, OCP = oral contraceptive pill

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the current update on morphomolecular classifications of HCA, HCC, and iCCA.
- Identify the genetic abnormalities and associated imaging features of select HCAs, HCCs, and iCCAs.
- Discuss management and prognostic implications of morphomolecular classifications of HCA, HCC, and iCCA.

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Hepatocellular adenomas (HCAs), hepatocellular carcinomas (HCCs), and intrahepatic cholangiocarcinomas (iCCAs) are a highly heterogeneous group of liver tumors with diverse pathomolecular features and prognoses. High-throughput gene sequencing techniques have allowed discovery of distinct genetic and molecular underpinnings of these tumors and identified distinct subtypes that demonstrate varied clinicobiologic behaviors, imaging findings, and complications. The combination of histopathologic findings and molecular profiling form the basis for the morphomolecular classification of liver tumors. Distinct HCA subtypes with characteristic imaging findings and complications include *HNF1A*–inactivated, inflammatory, β-catenin–activated, β-catenin–activated inflammatory, and sonic hedgehog HCAs. HCCs can be grouped into proliferative and nonproliferative subtypes. Proliferative HCCs include macrotrabecular-massive, *TP53*-mutated, scirrhous, clear cell, fibrolamellar, and sarcomatoid HCCs and combined HCC-cholangiocarcinoma. Steatohepatitic and β-catenin–mutated HCCs constitute the nonproliferative subtypes. iCCAs are classified as small-duct and large-duct types on the basis of the level of bile duct involvement, with significant differences in pathogenesis, molecular signatures, imaging findings, and biologic behaviors. Cross-sectional imaging modalities, including multiphase CT and multiparametric MRI, play an essential role in diagnosis, staging, treatment response assessment, and surveillance. Select imaging phenotypes can be correlated with genetic abnormalities, and identification of surrogate imaging markers may help avoid genetic testing. Improved understanding of morphomolecular features of liver tumors has opened new areas of research in the targeted therapeutics and management guidelines. The purpose of this article is to review imaging findings of select morphomolecular subtypes of HCAs, HCCs, and iCCAs and discuss therapeutic and prognostic implications.

Online supplemental material is available for this article.

Introduction

Recent advances in pathology, genetics, and oncology have resulted in a better understanding of the pathobiologic diversity of liver neoplasms, leading to a significant impact on management and prognostication. Adult hepatocytes are hypothesized to give rise to a diverse spectrum of liver tumors via direct transformation, dedifferentiation (into precursor/progenitor cells), or transdifferentiation (biliary-lineage cells). Hepatocellular adenomas (HCAs), hepatocellular carcinomas (HCCs), and intrahepatic cholangiocarcinomas (iCCAs) are highly heterogeneous at the cellular, genomic, and molecular levels (1–3).

TEACHING POINTS

- Focal nodular hyperplasia is the most common liver lesion demonstrating iso- to hyperintensity during the HBP. However, hyperintensity during the HBP may be seen in ß-catenin–mutated HCAs, unclassified HCAs, IHCAs, malignant HCAs, and well-differentiated HCCs. Combining the information obtained from conventional pre- and postcontrast MRI examinations and HBP signal intensity is crucial for accurate characterization of focal liver masses.
- Discontinuation of oral contraceptive pills and/or weight reduction and repeat MRI in 6 months is the first step in management of HCAs in women. This management strategy may allow regression of HCAs, permitting serial monitoring and conservative management in some patients. MRI surveillance every 6 months for 12–24 months is suggested if the lesions show interval regression or are smaller than 5 cm at 6-month follow-up MRI, with annual or biennial MRI thereafter depending on the risk factors and interval resolution. Biopsy and subsequent surgical resection may be considered for tumors that continue to grow or are larger than 5 cm. In women with HCAs less than 5 cm, treatment also depends on molecular subtyping.
- On the basis of morphomolecular taxonomic systems, HCCs are classified into two broad categories: proliferative (more aggressive) and nonproliferative (less aggressive). While nonproliferative HCCs carry *CTNNB1* or *JAK/STAT* pathway mutations and are well-differentiated tumors with good prognosis, proliferative HCCs are moderately to poorly differentiated tumors with *TP53* mutations and vascular endothelial growth factor (VEGF) activation, are more common in the setting of chronic hepatitis B infection, and have a worse prognosis. In addition to pathologic features, molecular test results and imaging findings also play a pivotal role in diagnosing these distinct subtypes and predicting the prognosis.
- Larger size, multiple lesions, substantial necrosis, infiltrating growth pattern, macrovascular thrombus, satellite nodules, internal arteries, irregular rimlike arterial phase hyperenhancement, lower apparent diffusion coefficient (ADC) values, peritumoral hypointensity during the HBP, and bile duct invasion suggest the possibility of a proliferative HCC and are predictive of poor overall survival. The presence of intracellular fat in HCCs is considered a favorable prognostic factor.
- Rhee et al proposed a subclassification of MF iCCAs into parenchymal and ductal types on the basis of MRI findings of bile duct abnormalities, as these two types significantly differ in biologic behavior and prognosis. The presence of bile duct dilatation and periductal tumor spread characterizes ductal-type MF iCCAs; these findings are notably absent in parenchymal-type tumors.

Evolving next-generation sequencing technologies have unraveled molecular and genomic characterization of liver tumors and identified distinct tumor subtypes with characteristic clinicobiologic behaviors and complications (Table 1) (1–3). The combination of histopathology, immunohistochemistry, and gene expression profiling form the basis for morphomolecular classification of liver tumors. This taxonomic schema of HCAs, HCCs, and iCCAs was included in the fifth edition of the World Health Organization (WHO) classification of digestive system tumors, published in 2019 (4).

Imaging plays a central role in detection, characterization, staging, and follow-up of focal liver lesions. Characteristic imaging findings are being identified that correlate with genetic mutations, the field of radiogenomics. Noninvasive imaging techniques and surrogate imaging markers allow differentiation of diverse liver tumors. On the basis of the discovery of novel genetic mutations and tumor pathways, new research areas involving pathway-specific targeted therapeutics are being pursued. Owing to improved understanding of pathogenesis, natural history, and prognosis of liver tumors, new treatment and follow-up guidelines are being developed by societies, including the American Association for the Study of Liver Diseases, National Comprehensive Cancer Network, and Organ Procurement and Transplant Network in association with the American College of Radiology (1,3,5).

In this article, we discuss the current updates on morphomolecular classifications of HCAs, HCCs, and iCCAs, with emphasis on genetic abnormalities and imaging findings and implications for management and prognosis.

Role of Imaging

Cross-sectional imaging techniques, including US, contrast-enhanced US, CT, and MRI, are commonly used for diagnosis and surveillance of focal liver lesions. Although CT is helpful in characterizing focal liver lesions, multiparametric MRI, combining conventional sequences with postcontrast and diffusion-weighted sequences, is the preferred modality for noninvasive diagnosis and characterization of focal liver masses (6).

Gadoxetic acid, a hepatobiliary-specific contrast agent used for enhanced MRI, is increasingly used as a problem-solving tool that allows assessment of indeterminate liver masses based on hepatocyte function and biliary excretion (7). Delayed hepatobiliary phase (HBP) images are typically obtained 20 minutes after contrast agent administration. The signal intensity of the focal mass is then compared with that of background liver parenchyma (hypointense vs isointense vs hyperintense), which helps in tumor characterization (7,8). While liver lesions without functioning hepatocytes are typically hypointense on HBP images, hyperintensity of focal lesions can be attributed to one of the following mechanisms: uptake by functioning hepatocytes, increased uptake by tumor cells, retention in extracellular space because of desmoplasia, peritumoral retention due to hyperplastic hepatocytes, and biliary tumor enhancement (7).

Focal nodular hyperplasia is the most common liver lesion demonstrating iso- to hyperintensity during the HBP. However, hyperintensity

Table 1: Genotypic Characteristics of Major Subtypes of HCA, HCC, and iCCA

Hepatic Neoplasm	Genetic Abnormalities
HCA	
HNF1A–inactivated HCA	Biallelic inactivating mutations of *HNF1A* and *MODY3*
Inflammatory HCA	Mutations in the *IL-6/JAK/STAT* pathway genes
β-catenin–activated HCA	β-catenin exon 3 (classic) and exon 7/8 mutations
β-catenin–activated inflammatory HCA	β-catenin mutation and activated Wnt signaling pathway
Sonic hedgehog HCA	Mutations of *INHBE* gene with activation of sonic hedgehog pathway
HCC	
Steatohepatitic HCC	Activation of *IL-6/JAK/STAT* pathway
β-catenin–mutated HCC	Mutations of *CTNNB1* gene and Wnt/β-catenin signaling pathway
Macrotrabecular-massive HCC	*TP53* mutations and amplification of *FGF19*
Scirrhous HCC	*TSC1* and *TSC2* mutations and TGF-β signaling activation
Fibrolamellar HCC	Microdeletion on chromosome 19 with *DNAJB1-PRKACA* translocation
iCCA	
Small-duct type	*IDH1* and/or *IDH2* mutations, *FGFR2* rearrangements, and *BRAF* mutations
Large-duct type	*KRAS* and *TP53* mutations, *COX-2* (primary sclerosing cholangitis [PSC] associated), *GNAS* mutations (liver fluke–associated), and *KRAS* mutations (PSC and hepatolithiasis associated)

during the HBP may be seen in β-catenin–mutated HCAs, unclassified HCAs, inflammatory HCAs (IHCAs), malignant HCAs, and well-differentiated HCCs (Figs E1, E2) (7,9). Combining the information obtained from conventional pre- and postcontrast MRI examinations and HBP signal intensity is crucial for accurate characterization of focal liver masses (Figs E1, E2) (7,8,10).

Hepatocellular Adenoma

HCAs comprise benign clonal proliferation of mature hepatocytes that primarily occur in noncirrhotic livers (11). *Hepatic adenomatosis* refers to the presence of more than 10 HCAs (11). Risk factors for HCAs include use of oral contraceptive pills (OCPs), steatohepatitis, excess androgen exposure, and hereditary metabolic disorders (11).

The classification of HCAs has been evolving owing to identification of mutations involving hepatocyte nuclear factor 1 α *(HNF1A)*, *IL6/JAK/STAT3* pathway genes, β-catenin (*CTNNB1*), and immunohistochemical markers such as liver fatty acid–binding protein, β-catenin, glutamine synthetase, C-reactive protein, and serum amyloid A, along with characteristic histopathologic findings (1). HCA is now considered a heterogeneous entity classified into specific subtypes on the basis of unique molecular signatures and histopathologic features. The current WHO classification recognizes the following HCA subtypes: *HNF1A*-inactivated, inflammatory, β-catenin–activated, β-catenin–activated inflammatory, sonic hedgehog, and unclassified HCAs (4,12). Several new subtypes of HCAs are identified on the basis of cause, pathobiology, and genetic features (12).

On the basis of characteristic MRI findings, specific HCA subtypes may be identified noninvasively; radiologists and pathologists play important roles in classifying morphomolecular subtypes of HCA (Table 2) (13). Familial hepatic adenomatosis has been linked to *HNF1A* germline mutations and maturity-onset diabetes of the young type 3 (MODY3) (1). Other genetic diseases associated with HCAs include McCune-Albright syndrome with IHCA and type 1 glycogen storage disease HCAs (Fig 1) (1).

HCAs are mostly asymptomatic. However, hemorrhage (20%–25% of cases) and malignant transformation (5%–8% of cases) are worrisome complications and can be predicted on the basis of size and subtype (14). IHCAs and sonic hedgehog HCAs have an increased risk of spontaneous bleeding (14). The risk factors for malignant transformation are male sex, size greater than 5 cm, β-catenin mutations, and *TERT* promoter mutation (1,11).

HNF1A-inactivated HCA

HNF1A-inactivated HCAs (HHCAs; 30%–35% of HCAs) are usually sporadic. A minor subset of HHCAs is associated with germline mutations of the *HNF1A* gene and *MODY3* (1). HHCAs demonstrate lobulated contours and diffuse steatosis at pathologic analysis; the defining feature is biallelic inactivation of *HNF1A* and loss of *LFABP* expression at immunohistochemical analysis (1).

Table 2: Clinical, Pathologic, and MRI Characteristics of Major Subtypes of HCA

HCA Subtype	Salient Clinical and Pathologic Features	Salient MRI Characteristics
HNF1A-inactivated HCA	Lobulated appearance and diffuse steatosis at pathologic analysis Low risk of complications	Diffuse and uniform signal loss on opposed-phase T1-weighted gradient-echo images Hypointense in HBP with gadoxetic acid–enhanced MRI
Inflammatory HCA	OCP use, obesity, and fatty liver disease Sinusoidal dilatation, inflammatory infiltrates, and thick arteries at pathologic analysis Increased risk of spontaneous rupture and malignancy	Atoll sign: complete T2 peripheral hyperintense rim, enhancing in arterial phase with persistent portal venous/delayed enhancement, sinusoidal dilatation Crescent sign: partial T2-hyperintense peripheral rim Up to 30% are iso- to hyperintense in HBP—estimate the risk of malignancy
β-catenin–activated HCA	Men with androgen use Cytologic atypia, pseudoglands, cholestasis, hemorrhage, or necrosis Highest risk of malignancy in exon 3 β-catenin–activated subtype	No specific MRI features; heterogeneous owing to hemorrhage and necrosis Up to 80% are iso- to hyperintense in HBP—high risk of malignancy
β-catenin–activated inflammatory HCA	Pathologic features of both inflammatory and β-catenin–activated types Risk of malignancy and rupture	MRI features of both inflammatory and β-catenin–activated HCAs, depending on the components
Sonic hedgehog HCA	OCP use and obesity Argininosuccinate synthase overexpression at immunohistochemistry Highest risk for bleeding among all HCAs	Heterogenous hyperintense signal on precontrast T1-weighted images owing to hemorrhage hypointensity Commonly hypointense in HBP

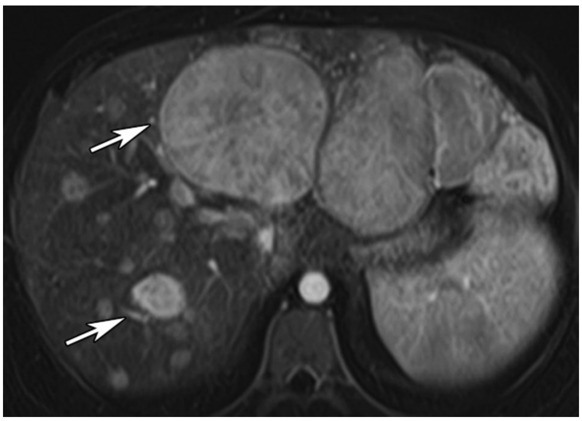


Figure 1. Hepatic adenomatosis in a 24-year-old man with glycogen storage disease type 1. Axial contrast-enhanced arterial phase T1-weighted MR image of the liver shows multiple homogeneously enhancing focal liver lesions (arrows). The left lobe masses were proved to be glycogen storage disease–associated unclassified HCAs at pathologic analysis.

At MRI, HHCAs demonstrate diffuse signal loss on opposed-phase T1-weighted gradient-echo (GRE) images, consistent with intracellular fat due to lipogenesis promoted by the production of nonfunctioning HNF1A protein (1,15). HHCAs are iso- to hyperintense on T2-weighted images, have moderate hyperenhancement in the arterial phase with persistent enhancement in the portal venous and delayed phases, and are hypointense in the HBP with gadoxetic acid–enhanced MRI (Fig 2) (13,15). HHCAs have a low risk of hemorrhage and malignant transformation (1,15).

Inflammatory HCA

IHCAs (5%–40% of HCAs) show a strong correlation with use of OCPs, obesity, fatty liver disease, alcohol abuse, diabetes, and glycogen storage disease (1,11). Activation of the *IL6/JAK/STAT3* pathway is the hallmark of IHCAs owing to somatic activating mutations in multiple genes (16). At pathologic analysis, IHCAs show focal inflammatory infiltrates, sinusoidal dilatation, and thick arteries; overexpression of serum amyloid A and C-reactive protein are characteristic immunohistochemical findings (17).

At MRI, IHCAs typically show hyperintense signal on T2-weighted images, which can be diffuse or along the periphery, appearing as a band referred to as the atoll sign. They have arterial phase enhancement, which persists in the portal venous and delayed phases (Fig 3) (9,13,15,18). A peripheral T2-hyperintense partial rim with arterial phase enhancement that persists during the portal venous and delayed phases is known as the crescent sign, a newly described sign that indicates focal sinusoidal dilatation (9,15,18). IHCAs are usually diffusely hypointense on HBP

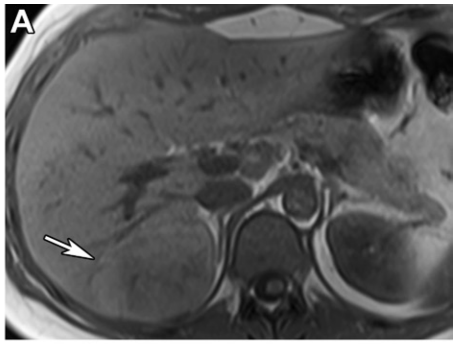
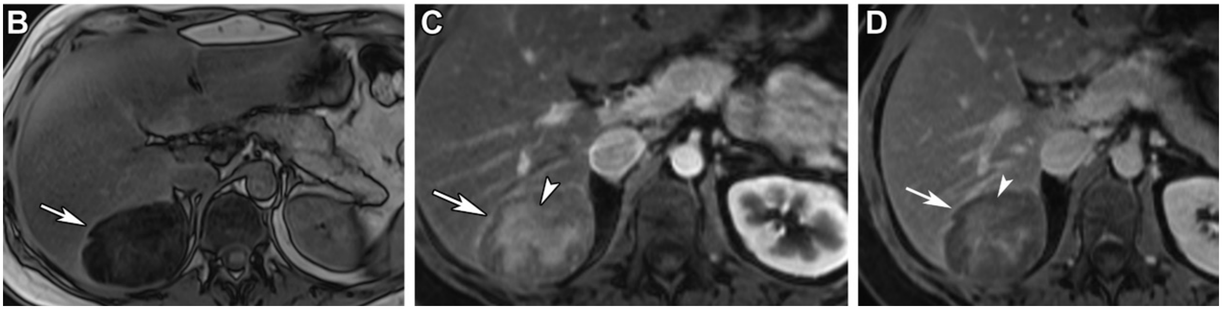


Figure 2. HHCA in a 40-year-old woman. **(A, B)** Axial in-phase **(A)** and opposed-phase **(B)** T1-weighted MR images of the liver show a well-circumscribed focal liver mass (arrow) with signal drop on the opposed-phase image **(B)**, diagnostic of intracellular fat. **(C, D)** Axial contrast-enhanced T1-weighted MR images in the arterial phase **(C)** and portal venous phase **(D)** show the lesion (arrow), which has mild heterogeneous internal enhancement (arrowhead). These MRI findings are consistent with HHCA.



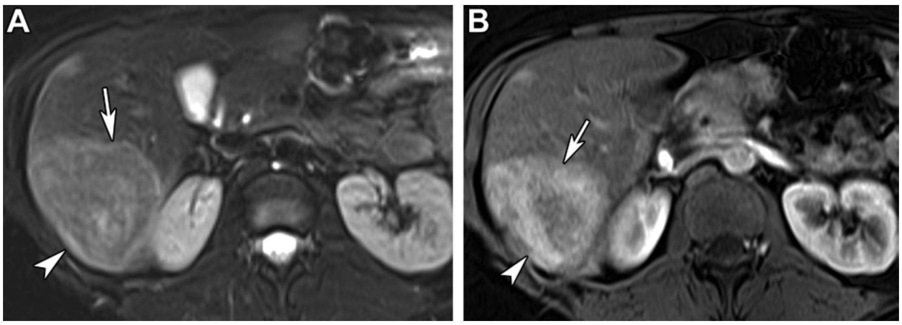


Figure 3. IHCA in a 38-year-old woman. **(A)** Axial T2-weighted MR image shows a moderately T2-hyperintense mass (arrow) with a hyperintense rim (arrowhead) in the right lobe of the liver. **(B)** Axial contrast-enhanced arterial phase T1-weighted MR image shows moderate heterogeneous enhancement within the mass (arrow), which has an avidly enhancing peripheral rim (arrowhead). These MRI findings are suggestive of an IHCA, which was proved at pathologic analysis.

images. However, up to 30% of them can appear isointense or hyperintense, which can be used to estimate the risk of malignancy, as there is increased β-catenin activation in this subgroup (Fig E1) (7,15). IHCAs carry an increased risk of spontaneous hemorrhage (1,11).

β-Catenin–activated HCA

β-catenin–activated HCAs (b-HCAs; 10% of HCAs) are further categorized into β-catenin exon 3 (classic) and exon 7/8 mutated HCAs, depending on the site of the mutations (16). b-HCAs demonstrate cytologic atypia, pseudoglands, cholestasis, hemorrhage, or necrosis (11,14). They display diffuse or focal glutamine synthetase expression and nuclear positivity for β-catenin at immunohistochemical analysis (11).

There are no specific imaging features for b-HCAs. The variable hyperintense signal can be seen on T1- and T2-weighted images, depending on intralesion hemorrhage or necrosis; they may demonstrate heterogeneous arterial phase enhancement (1,13,15). Select b-HCAs may demonstrate portal venous and/or delayed phase washout, with the delayed enhancing capsule mimicking HCC.

b-HCAs carry the highest risk of malignant transformation, which correlates with wingless-type integration site protein ligand (Wnt) signaling activation (11). Exon 3 mutations, which are more common in men with androgen use, confer a higher risk of malignant transformation (16). Up to 80% of b-HCAs demonstrate isointensity or hyperintensity during the HBP and are associated with a higher risk for malignancy (7,19). In their study, Reizine et al (20) showed that hepatospecific contrast agent uptake during the HBP is strongly associated with β-catenin activation in HCAs, thus predicting molecular subtype and risk of malignancy (20).

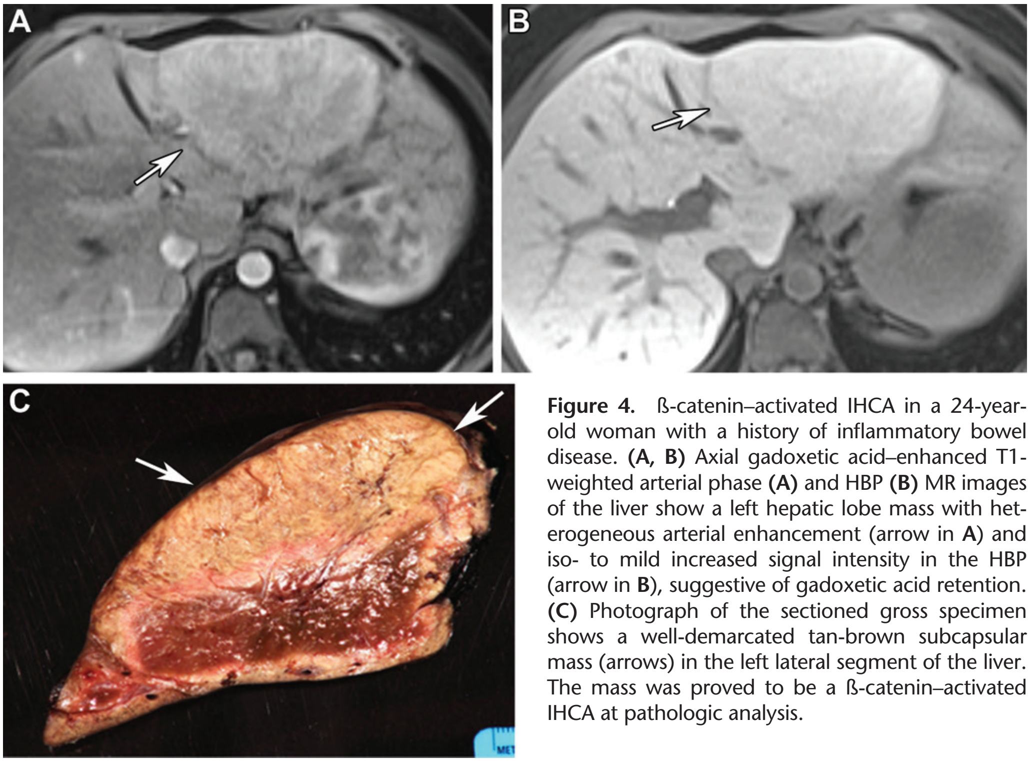


Figure 4. ß-catenin–activated IHCA in a 24-year-old woman with a history of inflammatory bowel disease. **(A, B)** Axial gadoxetic acid–enhanced T1-weighted arterial phase **(A)** and HBP **(B)** MR images of the liver show a left hepatic lobe mass with heterogeneous arterial enhancement (arrow in **A**) and iso- to mild increased signal intensity in the HBP (arrow in **B**), suggestive of gadoxetic acid retention. **(C)** Photograph of the sectioned gross specimen shows a well-demarcated tan-brown subcapsular mass (arrows) in the left lateral segment of the liver. The mass was proved to be a ß-catenin–activated IHCA at pathologic analysis.

β-Catenin–activated IHCA

β-catenin–activated IHCAs (10%–15% of HCAs) show phenotypic features of IHCAs along with β-catenin mutation and activated signaling pathway (1,12,14). β-catenin–activated IHCAs show pathologic features of both IHCA and β-catenin–activated HCA. There are no specific imaging features. While T2-hyperintense signal along with arterial phase hyperenhancement are suggestive of an inflammatory component, contrast agent retention during the delayed HBP signifies stronger β-catenin activation and higher risk for malignancy (Fig 4) (7,18).

Sonic Hedgehog HCA

Sonic hedgehog HCA (4% of HCAs) is a new molecular subtype characterized by activation of the sonic hedgehog pathway (associated with lipid metabolism and liver regeneration) due to mutations of the *INHBE* gene (12,21). Sonic hedgehog HCAs are often associated with OCP use and obesity and carry the highest risk for bleeding (12). Argininosuccinate synthase *(ASS1)* overexpression is a hallmark feature of sonic hedgehog HCA (21). Sonic headgehog HCAs commonly demonstrate hyperintense signal on precontrast T1-weighted images owing to hemorrhage and hypointensity in the HBP (1,15).

Unclassified HCAs and Novel Subtypes

Unclassified HCAs (5%–10% of HCAs) do not show distinctive pathogenetic features and are a diagnosis of exclusion (11). They may appear iso- to hyperintense during the HBP, similar to the appearance of focal nodular hyperplasia (Figs E1, E3).

Recently described novel HCA subtypes based on etiologic and histopathologic findings include androgen-related, pigmented (heavy lipofuscin deposition), myxoid, and glycogen storage disease adenomas (Fig 1) (11). Most androgen-related and pigmented adenomas demonstrate β-catenin–activation cytologic atypia and carry a high risk of malignant transformation (1,11). However, further studies are needed to understand the pathogenesis and incidence of malignancy in androgen-related and pigmented HCAs.

Management of HCAs

HCAs are managed mainly on the basis of patient sex, tumor size, subtype, and associated complications (Fig 5) (1,22,23). Morphologic-genotypic correlation and potential risk of complications among HCAs were developed principally from estrogen-related HCAs and may not always fit well for HCAs developing without the estrogen influence (11). However, they are presumed to be similar, but this remains less clearly proven (11).

Owing to the high risk of malignant transformation, surgical resection is recommended for all HCAs in men and exon 3 mutated b-HCAs (1,22,23). Discontinuation of OCPs and/or weight reduction and repeat MRI in 6 months is the first step in management of HCAs in women. This management strategy may allow regression

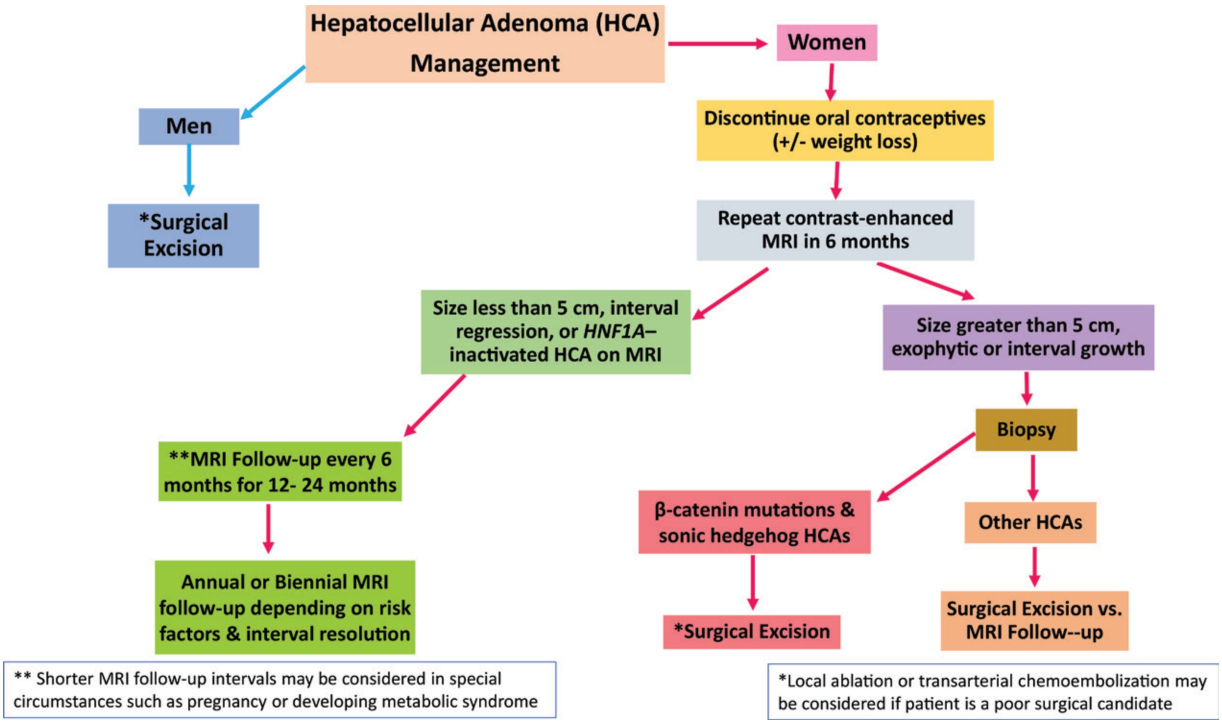


Figure 5. Flowchart illustrates management of HCAs, which mainly depends on the patient's sex, the size of the mass, and morphomolecular characterization with MRI and/or pathologic findings.

of HCAs, permitting serial monitoring and conservative management in some patients.

MRI surveillance every 6 months for 12–24 months is suggested if the lesions show interval regression or are smaller than 5 cm at 6-month follow-up MRI, with annual or biennial MRI thereafter depending on the risk factors and interval resolution. Biopsy and subsequent surgical resection may be considered for tumors that continue to grow or are larger than 5 cm. In women with HCAs smaller than 5 cm, treatment also depends on molecular subtyping (1,22,23).

Screening for *HNF1A* germline mutations and type I glycogen storage disease should be performed in select patients depending on the risk factors (1,11).

Hepatocellular Carcinoma

HCCs (75%–85% of primary liver cancers) usually occur in patients with cirrhosis, commonly from hepatitis B or C infection, alcoholism, and nonalcoholic fatty liver disease (NAFLD). Up to 20% of HCCs may develop in the noncirrhotic liver (24). HCC is a highly heterogeneous malignancy, at both the molecular and histologic levels, with varied biologic behavior that results from multiple gene mutations and epigenetic alterations (5,24).

The three commonly mutated genes in HCC include *TERT*, β-catenin, and *TP53* (5,24). The genetic mutations affect multiple signaling pathways, such as telomerase maintenance, cell cycle regulation, β-catenin/Wnt signaling, and the mammalian target of rapamycin (mTOR) and mitogen-activated protein kinase (MAPK) pathways that drive carcinogenesis (5,24). Distinct mutations and signaling pathways are associated with HCCs arising from different causes, resulting in distinct HCC genotypes/phenotypes with characteristic molecular signatures and clinicobiologic behavior (5,24). Currently, a percutaneous biopsy of a focal liver lesion in the cirrhotic liver that demonstrates characteristic imaging features of HCC is not required for treatment, although the desire for molecular characterization or histologic grading or a clinical trial requirement are the potential indications for biopsy.

On the basis of morphomolecular taxonomic systems, HCCs are classified into two broad categories: proliferative (more aggressive) and nonproliferative (less aggressive) (5,25). While nonproliferative HCCs carry *CTNNB1* or *JAK/STAT* pathway mutations and are well-differentiated tumors with good prognosis, proliferative HCCs are moderately to poorly differentiated tumors with *TP53* mutations and vascular endothelial growth factor (VEGF) activation, are more common in the setting of chronic hepatitis B infection, and have a worse prognosis (5,24,25). In addition to pathologic features, molecular test results and imaging findings also play a pivotal role in diagnosing these distinct subtypes and predicting the prognosis (26,27).

Table 3: Clinical, Pathologic, and CT and MRI Characteristics of Select Subtypes of HCC

HCC Subtype	Salient Clinical and Pathologic Features	Salient CT and MRI Characteristics
Steatohepatitic HCC	Metabolic syndrome, hepatic steatosis Steatohepatitis at pathologic analysis Good prognosis	T2-hyperintense mass, diffuse loss of signal on opposed-phase T1-weighted images owing to intracellular fat
β-catenin–mutated HCC	Well-differentiated tumor cells in microtrabecular and pseudoglandular pattern, and cholestasis Good prognosis	Iso- to hyperintense in HBP with gadoxetic acid–enhanced MRI
MTM HCC	Macrotrabecular growth pattern in >50% of tumor and frequent vascular invasion High serum AFP level and worse prognosis	Large heterogeneous tumor with substantial necrosis, rimlike arterial phase enhancement, and intratumoral arteries and peritumoral arterial enhancement
Scirrhous HCC	Dense intratumoral fibrosis in >50% of tumor Decreased survival compared with that of conventional HCC	Mimics imaging features of iCCA at imaging Hypoattenuating and T2-hypointense mass with peripheral rim and heterogeneous arterial phase enhancement
Clear cell HCC	Clear cells at pathologic analysis owing to glycogen accumulation Good prognosis	Mimics imaging features of conventional HCC
Fibrolamellar HCC	Young patient without background liver disease; normal AFP level Polygonal or spindle-shaped tumor cells, separated by lamellar sheetlike fibrous bands	Heterogeneously enhancing mass with a large central scar and calcifications MRI: iso- to hypointense at T1-weighted imaging and slightly hyperintense at T2-weighted imaging, with a hypointense central scar

Note.—AFP = α-fetoprotein, MTM = macrotrabecular massive.

In addition to conventional HCC, HCCs are categorized into eight distinct subtypes per the 2019 revised World Health Organization (WHO) classification system (Table 3) (28). Up to 35% of HCCs can be classified under one of these subtypes (29,30). Select HCC subtypes with characteristic imaging findings are discussed in the following sections.

Steatohepatitic HCC

Steatohepatitic HCC (5%–20% of HCCs) is characterized by intratumoral fat deposition (30,31). Most steatohepatitic HCCs occur in patients with metabolic syndrome or alcoholic liver disease, with the background liver also showing steatohepatitis (29). Characteristic histopathologic features of steatohepatitis, including steatosis, inflammation, and typical "chicken-wire pattern" pericellular fibrosis, are seen (31). Activation of the *IL-6/JAK/STAT* pathway and low frequency of *CTNNB1, TERT*, and *TP53* mutations are common in HCC (29).

At imaging, steatohepatitic HCC appears as an echogenic mass at US and a hypoattenuating mass at noncontrast CT. It shows arterial phase enhancement with washout in the portal venous and delayed phases (32,33). At MRI, steatohepatitic HCCs have hyperintense signal on T2-weighted images and diffuse loss of signal on opposed-phase T1-weighted gradient-echo images owing to intracellular fat (Fig 6) (32). Although the prognosis of steatohepatitis HCC is similar to that of conventional HCC, satellite nodules and microvascular invasion are uncommon (31,32).

β-Catenin (*CTNNB1*)–mutated HCC

β-catenin (*CTNNB1* gene)–mutated HCCs (10%–20% of HCCs) show activating mutations of the *CTNNB1* gene, which encodes for β-catenin and resultant activation of the Wnt/β-catenin signaling pathway (29). β-catenin–mutated HCC shows well-differentiated tumor cells in microtrabecular and pseudoglandular patterns as well as cholestasis (2). Dysregulation of bile salt transporters such as OATP1B3 causes retention of contrast material in the HBP.

β-catenin–mutated HCCs may appear iso- to hyperintense in the HBP at gadoxetic acid–enhanced MRI, appear hypointense on diffusion-weighted images, and show higher apparent diffusion coefficient (ADC) values (Fig 7) (34). The presence of iso- to hyperintensity during the HBP is an imaging biomarker of OATP1B3 and Wnt/β-catenin activation and suggests the diagnosis of β-catenin–mutated HCC (35). This imaging feature can be used to predict the effect of immune checkpoint inhibitor therapy in unresectable HCCs and may act as a surrogate marker for this purpose (36).

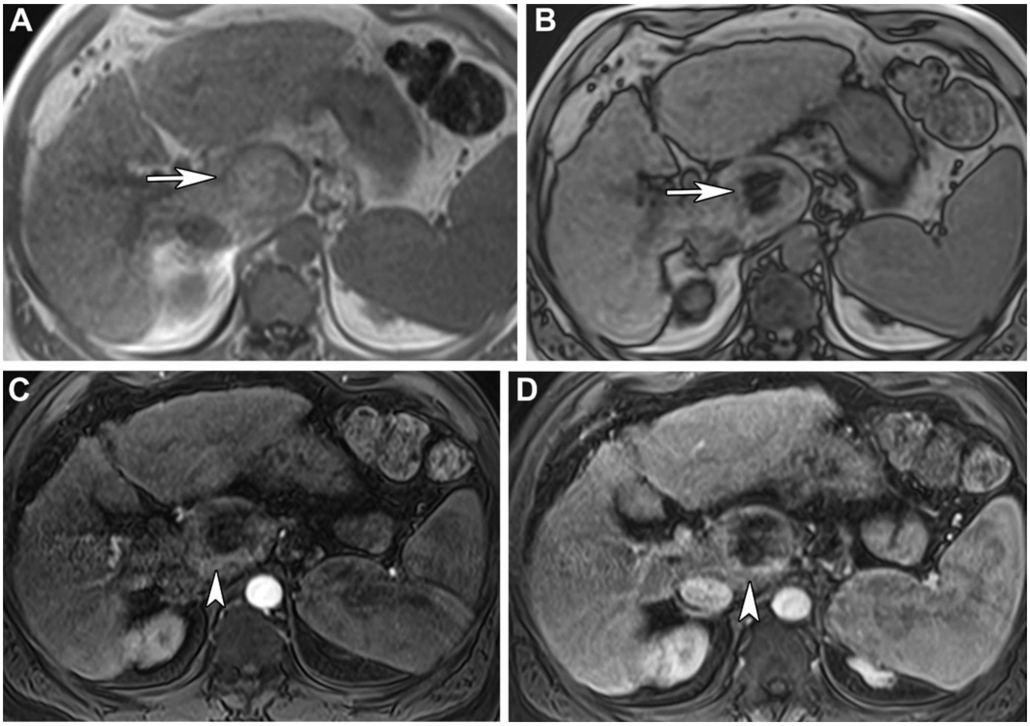


Figure 6. Steatohepatitic HCC in a 53-year-old man. **(A, B)** Axial in-phase **(A)** and opposed-phase **(B)** fat-saturated T1-weighted MR images show a well-circumscribed liver mass (arrow) in the caudate lobe with signal drop on the opposed-phase image, diagnostic of intracellular fat. **(C, D)** Axial contrast-enhanced arterial phase **(C)** and portal venous phase **(D)** fat-saturated T1-weighted MR images show irregular and heterogeneous peripheral enhancement of the mass (arrowhead). The mass was proved to be steatohepatitic HCC at pathologic analysis.

Figure 7. ß-catenin–mutated HCC in a 77-year-old man. Axial contrast-enhanced T1-weighted MR images of the liver during the arterial phase **(A)** and portal venous phase **(B)** show a right hepatic lobe mass (arrow) with heterogeneous arterial enhancement but no significant portal venous washout. The mass was confirmed to be ß-catenin–mutated HCC at pathologic analysis.

Macrotrabecular-Massive HCC

Macrotrabecular-massive (MTM) HCC (5%–10% of HCCs) is associated with large tumor size, tumor necrosis, frequent vascular invasion, high serum α-fetoprotein level (>100 ng/mL [100 μg/L]), aggressive biologic behavior, and poor prognosis (29,30,37). MTM HCCs are highly proliferative tumors commonly seen in patients with hepatitis B virus infection and in Barcelona Clinic Liver Cancer stage B and C liver disease (37). At pathologic analysis, MTM HCCs are often poorly differentiated and demonstrate characteristic prominent thick trabeculae (greater than six cells in thickness) involving at least 50% of the tumor (37). Frequent *TP53* mutations and *FGFR3* amplification and overexpression of *VEGFA* and *ANGPT2* are hallmark features of MTM HCC (29,37).

At imaging, MTM HCCs tend to be larger (average size > 5 cm) and heterogeneous with substantial necrosis, which appears as high signal intensity on T2-weighted images and lack of enhancement (Fig 8) (38). MTM HCCs may also exhibit rimlike arterial phase enhancement

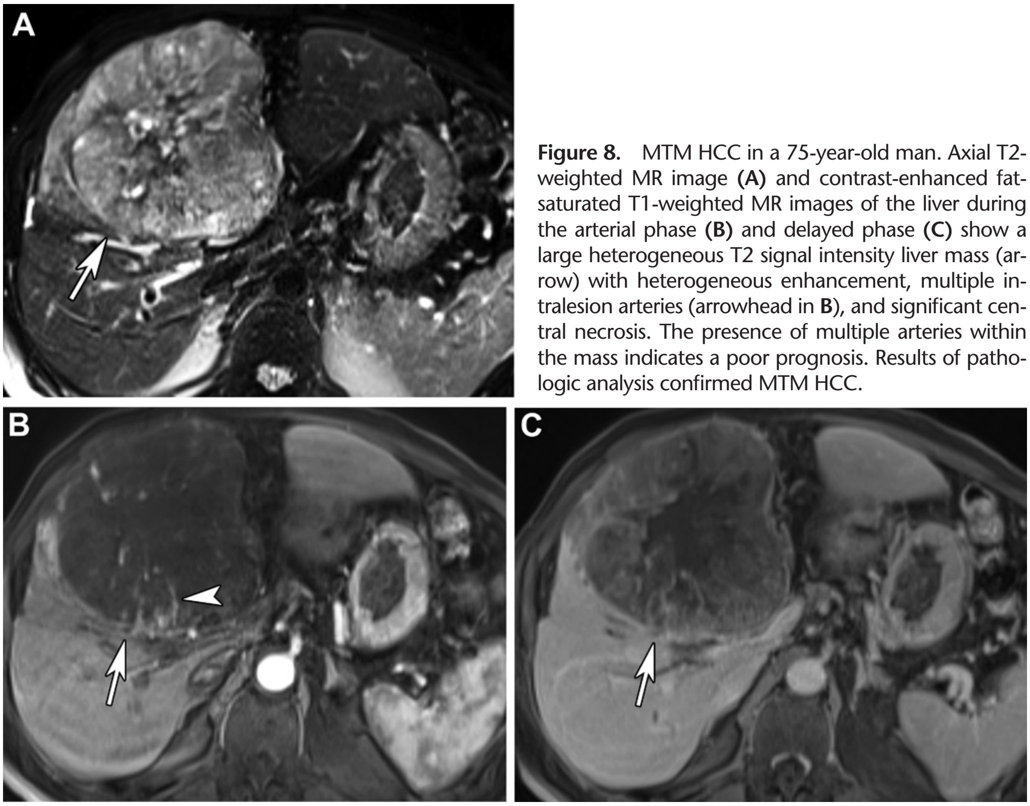


Figure 8. MTM HCC in a 75-year-old man. Axial T2-weighted MR image **(A)** and contrast-enhanced fat-saturated T1-weighted MR images of the liver during the arterial phase **(B)** and delayed phase **(C)** show a large heterogeneous T2 signal intensity liver mass (arrow) with heterogeneous enhancement, multiple intralesion arteries (arrowhead in **B**), and significant central necrosis. The presence of multiple arteries within the mass indicates a poor prognosis. Results of pathologic analysis confirmed MTM HCC.

reminiscent of CCA and other HCC subtypes (28). On gadoxetic acid–enhanced MR images, presence of more than 50% arterial phase hypovascular component, intratumoral arteries, peritumoral arterial enhancement, and nonsmooth tumor margins during the HBP suggest the possibility of MTM HCC (39).

Compared with conventional HCC, MTM HCCs demonstrate lower apparent diffusion coefficient (ADC) values in the solid portions of the tumor (38). MTM HCCs have a higher incidence of early recurrence and poor prognosis. Tumor size, necrosis, macrovascular invasion, and satellite nodules can help predict recurrence risk in MTM HCC (37).

Scirrhous HCC

Scirrhous HCCs (<5% of HCCs) are characterized by dense fibrous stroma occupying more than 50% of the unencapsulated tumor, in which cords of tumor cells resembling hepatic stem and progenitor cells are embedded (Fig 9) (40). They are characterized by *TSC1*/*TSC2* mutations with transforming growth factor–β signaling activation (29,40).

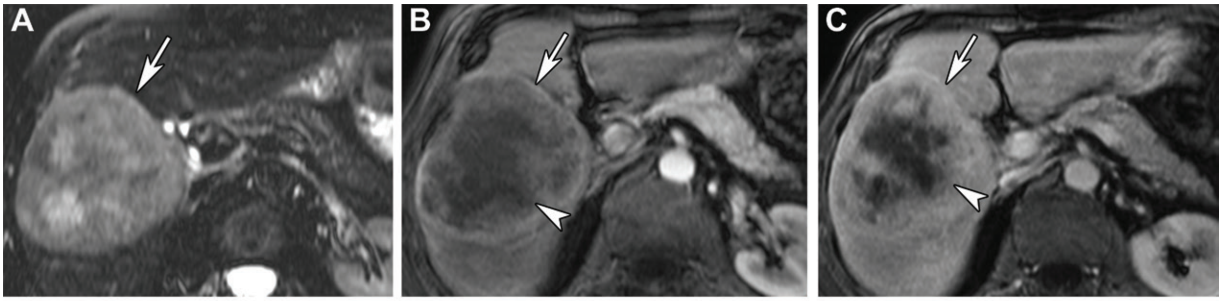
Scirrhous HCCs mimic iCCAs at imaging owing to significant desmoplasia. At CT and MRI, they appear as hypoattenuating and T2-hypointense masses, respectively, with peripheral rim and heterogeneous arterial phase and portal venous phase enhancement and progressive centripetal enhancement in the delayed phase (41). A typical targetoid appearance has been described on MR images, characterized by heterogeneous T2 hyperintensity with a central dark area and enhancing capsule on delayed phase images (Fig 9) (42,43). On gadoxetic acid–enhanced MR images, tumors typically show lobular margins with diffuse hypointensity during the HBP (42–44). Aggressive features such as infiltrative growth and vascular invasion frequently occur in large tumors, resulting in decreased survival compared with that of conventional HCCs (40).

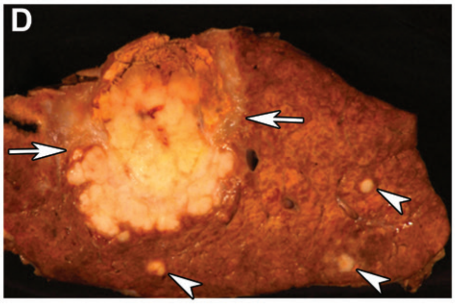
Clear Cell HCC

Clear cell HCCs (3%–7% of HCCs) are characterized by clear cytoplasm in more than 80% of the tumor owing to glycogen accumulation (31). At CT and MRI, clear cell HCCs show findings typical of conventional HCCs, including prompt enhancement in the arterial phase and washout in the portal venous and delayed phases (45). Clear cell HCC frequently has a pseudocapsule (Fig 10) (45). The prognosis of clear cell HCCs has been reported to be better than that of conventional HCCs (45).

Fibrolamellar HCC

Fibrolamellar HCC is a rare subtype that typically occurs in children and young adults without any gender predilection or known risk factors for HCC and with normal α-fetoprotein





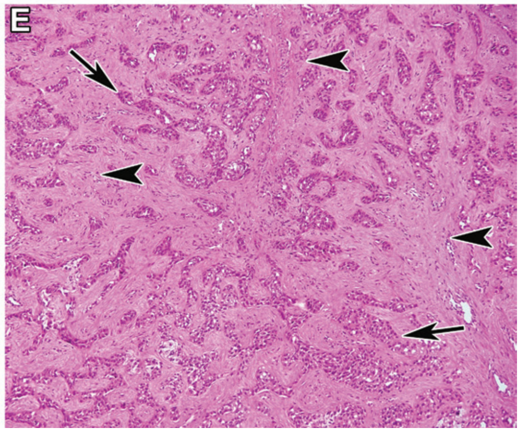


Figure 9. Scirrhous HCC in two patients. **(A–C)** Scirrhous HCC in a 72-year-old man with elevated liver enzyme levels. Axial T2-weighted **(A)** and contrast-enhanced fat-saturated arterial phase **(B)** and portal venous phase **(C)** T1-weighted MR images of the liver show a heterogeneously T2-hyperintense right hepatic lobe mass (arrow) with heterogeneous peripheral enhancement, predominantly during the portal venous phase, and central areas of necrosis (arrowhead in **B** and **C**). Results of pathologic analysis confirmed scirrhous HCC. **(D, E)** Scirrhous HCC in a 32-year-old man with a history of hepatitis B viral infection. **(D)** Photograph of the sectioned gross specimen shows a large tan-gray firm well-circumscribed mass (arrows) and three small similar-appearing nodules (arrowheads). **(E)** Photomicrograph shows tumor cells arranged in anastomosing nests, trabeculae, and cords (arrows) surrounded by abundant fibrocollagenous stroma (arrowheads), consistent with scirrhous HCC. (Hematoxylin-eosin stain; original magnification, ×10.)

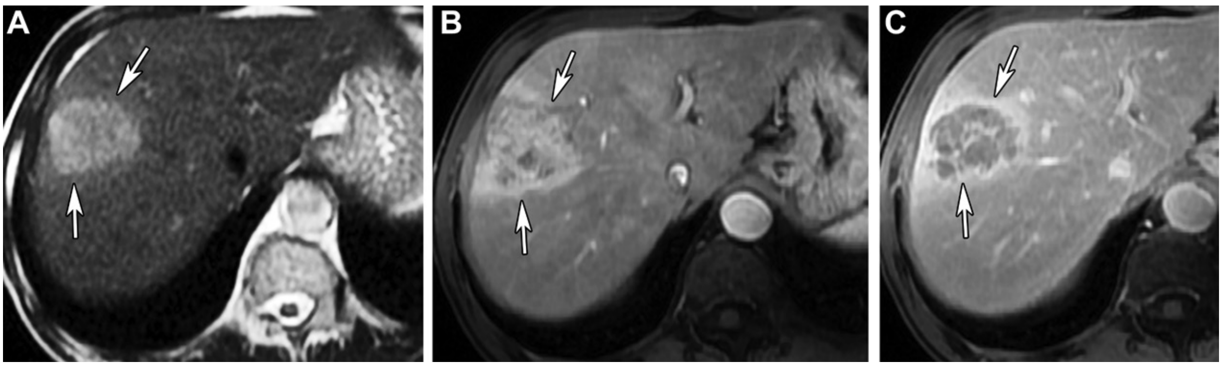


Figure 10. Clear cell HCC in a 55-year-old man. Axial T2-weighted **(A)** and contrast-enhanced fat-saturated arterial phase **(B)** and portal venous phase **(C)** T1-weighted MR images of the liver show a heterogeneously T2 hyperintense mass (arrows) in the right lobe of the liver, with heterogeneous enhancement during the arterial phase and central washout during the portal venous phase. The mass was proved to be clear cell HCC at pathologic analysis.

level (30). At pathologic analysis, fibrolamellar HCCs are well-circumscribed tumors with dense desmoplasia. Clusters of large polygonal or spindle-shaped tumor cells are arranged in cords, separated by lamellar sheetlike fibrous bands (Fig 11). Fibrolamellar HCCs show characteristic *DNAJB1–PRKACA* translocation owing to microdeletion on chromosome 19 (31).

On US images, a large mixed-echogenicity mass is depicted with a central echogenic scar (46). On CT images, fibrolamellar HCC appears as a well-circumscribed heterogeneously enhancing mass with a large central scar and calcifications (Fig 11) (46). On MR images, fibrolamellar HCCs are iso- to hypointense on T1-weighted images and slightly hyperintense to the liver on T2-weighted images, with a hypointense central scar. They demonstrate heterogeneous arterial phase hyperenhancement and become isointense or hypointense during the portal venous and delayed phases (Fig E4) (46).

A disproportionately large central scar (>2 cm) and calcifications are useful imaging features that help differentiate fibrolamellar HCCs from focal nodular hyperplasia (46). Unlike focal nodular hyperplasia, fibrolamellar HCCs do not retain contrast material in the HBP (Fig E4) (47). Surgical resection is the treatment of choice for fibrolamellar HCCs; patients have a better prognosis than

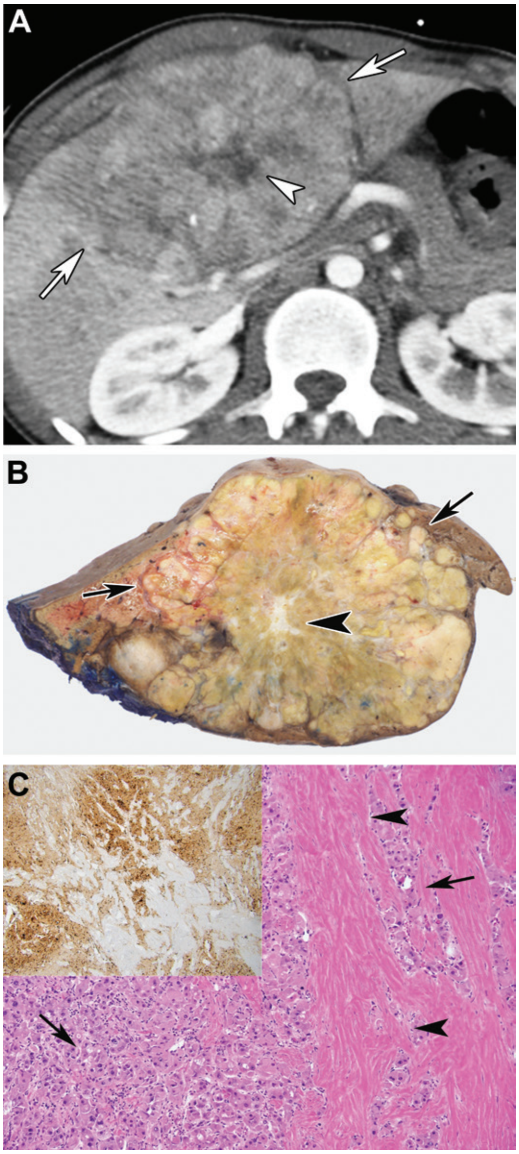


Figure 11. Fibrolamellar HCC in a 19-year-old man. **(A)** Axial contrast-enhanced arterial phase CT image of the liver shows a large left lobe liver mass with moderate enhancement with a "spoke-wheel pattern" (arrows) and a central hypoenhancing scar (arrowhead). **(B)** Photograph of the sectioned gross specimen shows a liver mass with tan-yellow cut surface and areas of green staining (arrows) with central fibrotic scar (arrowhead). **(C)** Photomicrograph shows neoplastic cells with abundant oncocytic cytoplasm (arrows) in the background of dense collagen bundles arranged in parallel lamellae (arrowheads). (Hematoxylin-eosin stain; original magnification, ×20). Inset image shows the tumor cells highlighted by arginase immunohistochemical stain. These findings are consistent with fibrolamellar HCC.

those with conventional HCCs in cirrhotic livers, whereas there is a somewhat similar prognosis in those with noncirrhotic livers (31).

Combined HCC-CCA

Combined HCC-CCA is an aggressive form of primary liver carcinoma, characterized by mixed phenotypic features of both hepatocytic and cholangiocytic differentiation within the same tumor and worse prognosis than that of conventional HCC (48). CT and MRI may demonstrate a combination of findings typical of conventional HCC, including arterial phase hyperenhancement with washout and a delayed enhancing pseudocapsule, as well as imaging features of iCCA, such as peripheral arterial phase hyperenhancement, progressive centripetal enhancement, and capsular retraction, within the same tumor (Fig 12) (49).

It is difficult to diagnose combined HCC-CCA preoperatively on the basis of imaging findings alone. Combining imaging features, elevated tumor marker levels, and pathologic findings helps clinch the diagnosis (48). At presentation, most patients are diagnosed with localized advanced or distant metastatic disease. The prognosis is dependent on the predominant component, being worst for those with a CCA component with fibrotic stroma (48,49).

Rare and New HCC Subtypes

Other rare variants of HCC include sarcomatoid, chromophobe, neutrophil-rich, and lymphocyte-rich HCCs (28); no specific imaging features have been described.

Recently, two new HCC subtypes with characteristic clinical, pathologic, and imaging findings have been described: fibronodular HCC and cytokeratin-19 (CK19) HCC (50,51). Fibronodular HCC is a low-grade tumor that is characterized by the presence of multiple nodules embedded in a fibrotic stroma and nonperipheral washout and has a good prognosis (50). CK19 HCC demonstrates significant diffusion restriction, arterial phase rim enhancement, and diffuse hypointensity in the HBP; it has aggressive behavior and a poor prognosis (27,51).

Differentiating Proliferative and Nonproliferative HCCs at Imaging

Select imaging biomarkers can help predict the aggressive behavior of HCCs and differentiate proliferative from nonproliferative HCC types (52,53). Larger size, multiple lesions, substantial necrosis, infiltrating growth pattern, macrovascular thrombus, satellite nodules, internal arteries, irregular rimlike arterial phase hyperenhancement, lower apparent diffusion coefficient (ADC) values, peritumoral hypointensity during the HBP, and bile duct invasion suggest the possibility of a proliferative HCC and are predictive of poor overall survival (Fig 13) (52,53). The presence of intracellular fat in an HCC is considered a favorable prognostic factor (Fig 6) (26).

Qualitative analysis of HBP tumor enhancement may help predict tumor aggressiveness. While nonproliferative HCCs are more likely

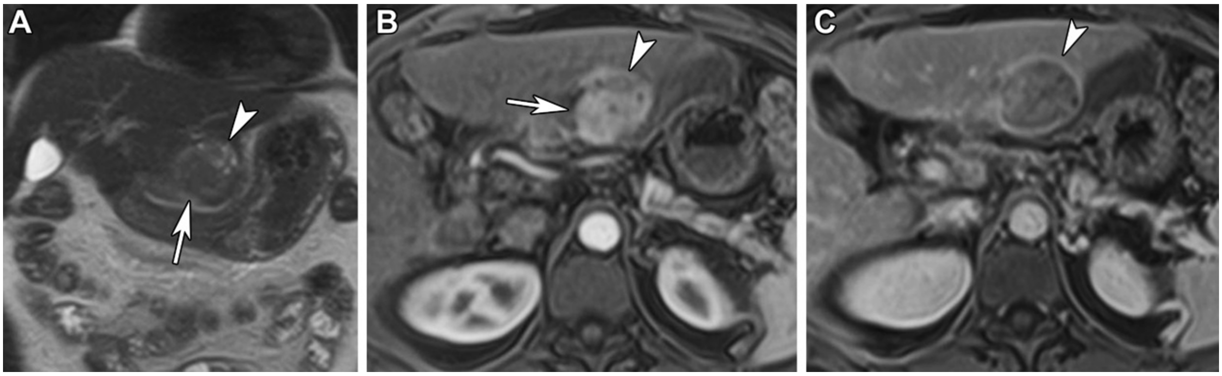


Figure 12. Combined HCC-CCA in a 60-year-old woman. **(A)** Coronal T2-weighted MR image of the liver shows a T2 iso- to mildly hyperintense left lobe mass (arrow) with heterogeneous T2 hyperintensities in its superior aspect (arrowhead). **(B)** Axial contrast-enhanced arterial phase fat-saturated T1-weighted MR image shows heterogeneous enhancement of the mass, with avid enhancement in its posterior aspect (arrow) and hypoenhancement anteriorly (arrowhead). **(C)** Axial contrast-enhanced delayed phase fat-saturated T1-weighted MR image shows washout and capsular enhancement of the mass (arrowhead). The mass was proved to be combined HCC-CCA at pathologic analysis.

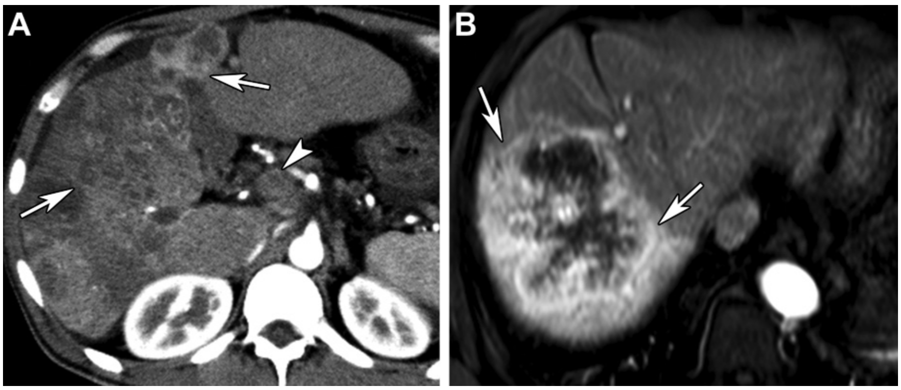


Figure 13. Imaging features of proliferative HCC with aggressive clinical behavior in two patients. **(A)** Infiltrating HCC in a 50-year-old man. Axial contrast-enhanced arterial phase CT image of the liver shows a large ill-defined heterogeneously enhancing mass (arrows) involving most of the right lobe and part of the left lobe, with tumor thrombus (arrowhead) in the main portal vein. **(B)** Poorly differentiated HCC in a 66-year-old man. Axial contrast-enhanced arterial phase fat-saturated T1-weighted MR image of the liver shows a heterogeneously enhancing mass in the right lobe with irregular rimlike peripheral enhancement (arrows).

to be hyperintense relative to the surrounding parenchyma in the HBP, aggressive HCCs with a poor prognosis are hypointense (Fig E2) (54). Dual-tracer PET/CT with fluorine 18 (^{18}F)–fluorodeoxyglucose (FDG) and ^{18}F-fluorocholine is also helpful in predicting the biologic behavior of an HCC, as proliferative HCCs show increased FDG avidity and nonproliferative HCCs are fluorocholine avid (54,55).

Management of HCCs

The overall prognosis of HCCs has improved, mainly owing to early diagnosis and available treatment options. Evolving morphomolecular classification of HCCs will have a significant impact on clinical outcomes and individualized treatment options based on molecular abnormalities (5,25). The Barcelona Clinic Liver Cancer staging system is commonly used to prognosticate and guide treatment strategies in HCC. This system considers tumor burden, liver function, and patient performance status for prognostication and selecting treatment options (24).

However, identifying new HCC molecular subtypes and imaging features for identifying them and predicting aggressiveness may influence treatment strategies in the future (5,25). For example, given the grim prognosis of MTM HCC, patients may benefit from adjuvant therapies and/or early liver transplantation after initial curative treatment. HCC driver genes such as *CTNNB1*, *TP53*, and *TERT* are considered main targets for novel therapeutic development (16). New immunotherapies (immune checkpoint inhibitors) and multikinase inhibitors are being developed to treat advanced HCCs (5,25).

Intrahepatic Cholangiocarcinoma

CCA, the second most common primary liver malignancy, comprises a diverse group of

Table 4: Clinical, Pathologic, and CT and MRI Characteristics of iCCAs

Characteristics	Small-Duct–Type iCCA	Large-Duct–Type iCCA
Location	Peripheral liver parenchyma	Proximal to the hepatic hilum
Risk factors	Nonbiliary cirrhosis, chronic viral hepatitis	Primary sclerosing cholangitis, liver fluke infection, hepatolithiasis
Precursor lesions	Unknown	Biliary intraepithelial neoplasia, intraductal papillary neoplasm
Histogenesis	Small bile ducts and ductules, hepatic progenitor cells	Intrahepatic large bile ducts and peribiliary glands
Pathologic features	MF growth pattern Tubular or ductular pattern with desmoplasia; non–mucin-secreting glands	Mainly periductal infiltrating Ductal or tubular pattern with desmoplasia; mucin-secreting glands
Molecular features	*IDH1* and/or *IDH2* and *FGFR2* mutations	*KRAS* and *TP53* mutations
Imaging features	Homogeneously hypoenhancing mass without associated biliary dilatation (parenchymal-type MF iCCA)	Liver mass with associated bile duct dilatation and periductal tumor spread (ductal-type MF iCCA; periductal infiltrating type iCCA)
Prognosis	Good prognosis	Worse prognosis

malignant biliary tract neoplasms with distinct anatomic distribution, epidemiologic risk factors, biologic behavior, and prognosis (3,56). CCAs are anatomically classified into three subtypes: intrahepatic, perihilar, and distal (3,56). While intrahepatic CCAs (iCCAs; 10%–20% of CCAs) arise above the second-order bile ducts, perihilar CCAs (50%–60% of CCAs) originate within the common hepatic duct and left or right hepatic duct or at their junction, and distal CCAs (20%–30% of CCAs) arise from the common bile duct distal to the cystic duct (3,56).

iCCAs (10%–15% of primary liver cancers) are adenocarcinomas that frequently demonstrate marked desmoplasia and invade portal lymphatics and venules. On the basis of the size and level of bile duct involvement, iCCAs are divided into small-duct types when they arise from septal and interlobular bile ducts and large-duct types when they originate within large central intrahepatic bile ducts (3,56). On the basis of their growth pattern, iCCAs can be classified into mass-forming (MF), periductal-infiltrating, intraductal-growing, and mixed types (56).

iCCAs may also be classified into two subsets on the basis of tumor genetics and pathways; while the more common proliferation type tumors (60%) are associated with activation of *RAS/MAPK* pathways, the inflammatory-type tumors demonstrate cholangiolar differentiation and activation of *IL6/STAT* pathways (56,57). In combination with histopathologic and immunohistochemical findings, new classification systems are being proposed for iCCAs dependent on the cell of origin and molecular abnormalities that may help develop targeted therapies.

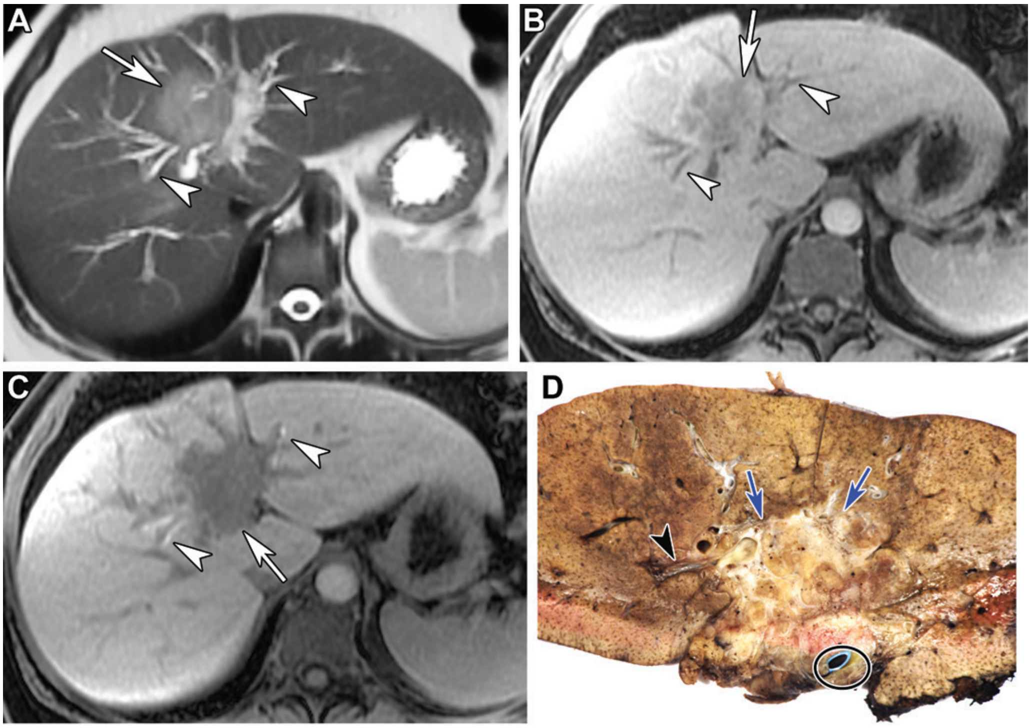
Small- and Large-Duct iCCA: Pathogenesis, Pathology, and Genetics

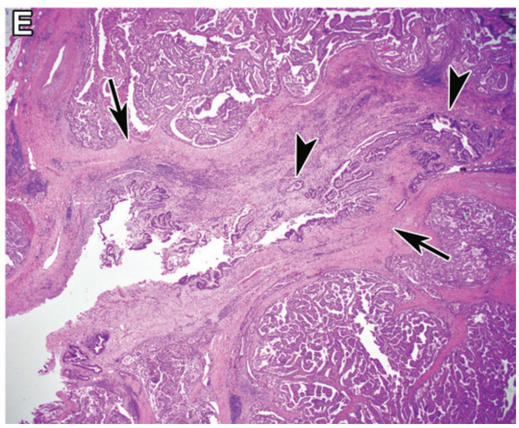
The cells of origin of small-duct iCCAs include cholangiocytes, stem cells in the canal of Hering, hepatic progenitor cells that have undergone transdifferentiation, and mature hepatocytes (58,59). The risk factors are similar to those of HCCs and include viral hepatitis (hepatitis C and B) infection and nonbiliary cirrhosis (3,58). An MF growth pattern with a white-grayish mass in the periphery of the liver is the most common appearance of small-duct iCCAs. It consists of small cuboidal to low columnar epithelium demonstrating proliferation of cordlike structures or ductular patterns with high cellularity and fibrous stroma in the center (58). *IDH1/IDH2* (isocitrate dehydrogenase) mutations (up to 20%), *FGFR2* (fibroblast growth factor receptor 2) rearrangements (about 15%), and *BRAF* mutations occur exclusively in small-duct iCCAs (Table 4) (3,56).

The cells of origin of large-duct iCCAs include bile duct epithelial cells or stem cells in the peribiliary glands surrounding the central intrahepatic bile ducts near the hilum (58,59). Biliary intraepithelial neoplasia and intraductal papillary neoplasm of the bile ducts are common premalignant lesions. The risk factors include hepatolithiasis, primary sclerosing cholangitis (PSC), choledochal cysts, and parasitic infestations (liver flukes) (3,58).

At gross pathologic analysis, large-duct iCCAs exhibit different growth patterns. At histologic analysis, mucin-secreting tall columnar epithelium with a papillary proliferation of tubular component and extensive desmoplastic stroma are common findings (Fig 14) (58). Bile duct wall thickening and sclerosis with tumoral obliteration of the large bile ducts are common findings.

Figure 14. MF iCCA arising from an intraductal papillary neoplasm of the bile duct in a 44-year-old woman. **(A–C)** Axial T2-weighted **(A)**, fat-saturated arterial phase **(B)**, and delayed HBP **(C)** gadoxetic acid–enhanced MR images show a T2-hyperintense left liver lobe mass (arrow) with heterogeneous peripheral arterial enhancement, which is hypointense on the delayed HBP image, and associated biliary dilatation (arrowheads). **(D)** Photograph of the sectioned gross specimen shows an ill-defined firm vaguely lobular liver mass with a tan-white to pale cut surface (arrows). The mass surrounds and compresses the deep hilar structures, and there is associated biliary dilatation (arrowhead). Also note the stent positioned in the bile duct (circle). **(E)** Photomicrograph shows intraductal papillary neoplasm of the bile duct with adjacent invasive CCA of the intrahepatic bile duct (arrows). Tumor shows papillary proliferation within the dilated bile duct (arrowheads). (Hematoxylin-eosin stain; original magnification, ×2.)





KRAS and *TP53* mutations are the most common genetic abnormalities in large-duct iCCAs. Other clinically relevant mutations reported in iCCAs include *SMAD4*, *BAP1*, *ARID1A*, *BRAF V600E* mutations, *HER2/neu* amplification, *BRCA1/2*, and *CDKN2A/B* (3,57). Large-duct iCCAs demonstrate increased expression of *COX-2* (PSC associated), *GNAS* mutations (liver fluke associated), and *KRAS* mutations (PSC and hepatolithiasis associated) (Table 4) (58). Genomic diversity and molecular heterogeneity of iCCAs are the basis of varied imaging findings and prognoses and dictate the need for personalized treatment options.

iCCA Updates in Imaging-based Classification

Dynamic multiphase contrast-enhanced CT, MRI, and MR cholangiopancreatography are the primary imaging methods for diagnosis and staging of iCCAs (60). The imaging findings correspond to the morphologic growth patterns of iCCAs (60,61).

MF iCCA typically appears as a well-defined intrahepatic mass with irregular margins, arterial phase peripheral rim enhancement, and progressive centripetal enhancement (60,61). Rhee et al (62) proposed a subclassification of MF iCCAs into parenchymal and ductal types on the basis of MRI findings of bile duct abnormalities, as these two types significantly differ in biologic behavior and prognosis (62). The presence of bile duct dilatation and periductal tumor spread characterizes ductal-type MF iCCAs; these findings are notably absent in parenchymal-type tumors (Figs 15, 16) (62). In their study, parenchymal-type MF iCCAs were associated

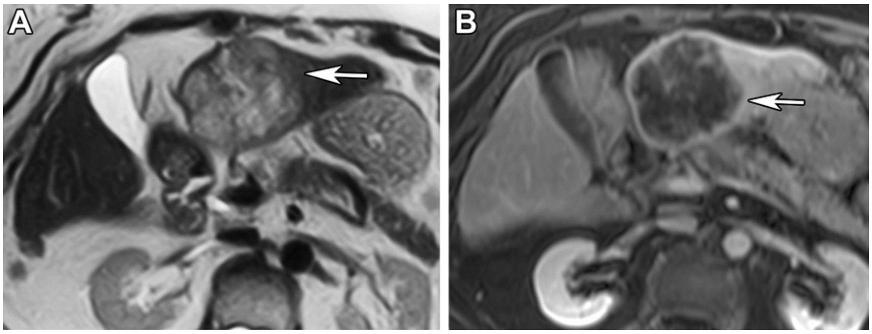


Figure 15. Parenchymal-type MF iCCA in an 81-year-old woman. Axial T2-weighted **(A)** and contrast-enhanced portal venous phase fat-saturated T1-weighted **(B)** MR images of the liver show a heterogeneous T2 signal intensity mass (arrow) in the left lobe of the liver with heterogeneous hypoenhancement in the portal venous phase. Note that there is no intrahepatic biliary dilatation. Results of pathologic analysis confirmed parenchymal-type MF iCCA.

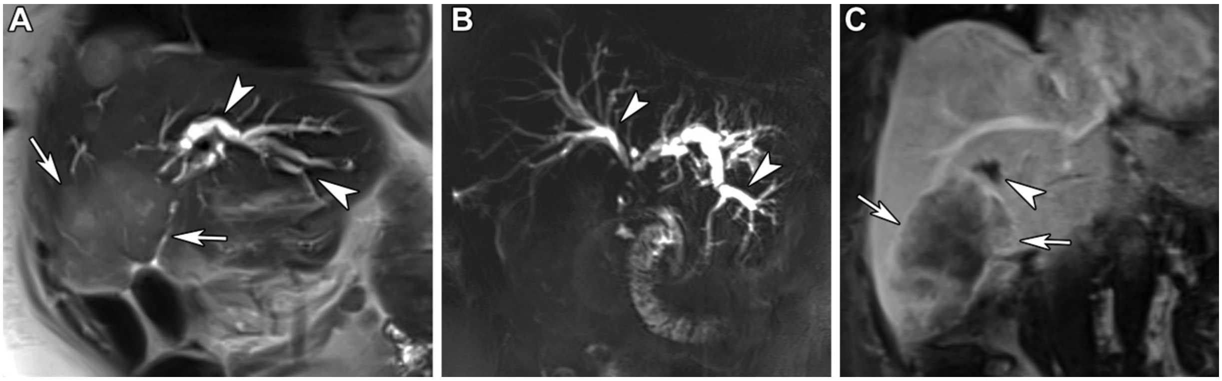


Figure 16. Ductal-type MF iCCA in a 79-year-old woman. Coronal T2-weighted MR image **(A)**, coronal T2-weighted two-dimensional MR cholangiopancreatogram **(B)**, and coronal contrast-enhanced portal venous phase fat-saturated T1-weighted MR image **(C)** of the liver show a moderately T2-hyperintense mass (arrows in **A**) in the right lobe of the liver, resulting in a biliary stricture at the bile duct confluence and associated intrahepatic biliary dilatation (arrowheads). The mass shows heterogeneous hypoenhancement during the portal venous phase (arrows in **C**) and was proved to be a ductal-type MF iCCA at pathologic analysis.

with chronic liver disease, small-duct–type histologic findings, and better prognosis.

The ductal-type MF iCCAs demonstrated large-duct histologic findings and were associated with high levels of serum tumor markers, large tumor size, frequent perineural invasion, lymph nodal disease, and overall poor prognosis (62). MF iCCAs demonstrate pseudo-washout at gadoxetic acid–enhanced MRI in the transitional phase secondary to background parenchyma demonstrating progressive enhancement (61). Diffusely hypo- or mixed hypointense signal or target appearance with a peripheral hypointense rim and central inhomogeneous cloudlike enhancement may be seen in the HBP, related to contrast material retention in the abundant fibrous stroma (8,61,63).

Periductal-infiltrating iCCAs and intraductal-growing iCCAs are less common and can coexist with MF iCCAs, resulting in a mixed pattern. Periductal-infiltrating iCCAs manifest as irregular periductal enhancing soft tissue associated with biliary stricture and upstream biliary dilatation (60,61). Intraductal-growing iCCA is the least common type and has the best prognosis. It commonly arises from intraductal papillary neoplasms of the bile ducts and appears as a polypoid intraluminal mass in the bile ducts (Fig 17). Alternatively, micropapillary lesions may not be visible at imaging but can be associated with variable segmental or lobar biliary dilatation. Additional imaging findings of iCCAs include satellite nodules, regional lymphadenopathy, and intrahepatic and systemic metastases (60).

Role of Imaging in Predicting Molecular Signatures and Prognosis

Some characteristic imaging features of iCCAs appear to correlate with genetic mutations, molecular signatures, and prognosis. Zhu et al (64) showed that *IDH1*-mutated iCCAs demonstrate different contrast-enhanced CT features than *IDH1*-wild iCCAs. *IDH1*-mutated iCCAs demonstrate intratumoral arteries with rim and internal enhancement during the arterial phase, whereas *IDH1*-wild tumors show diffuse or no enhancement without any intratumoral arteries (64). iCCAs with *IDH1* mutations demonstrate more enhancement during the arterial and portal venous phases, with significantly higher attenuation values and enhancement ratios (tumor attenuation/aortic attenuation) when compared with those of *IDH1*-wild iCCAs (Fig 18).

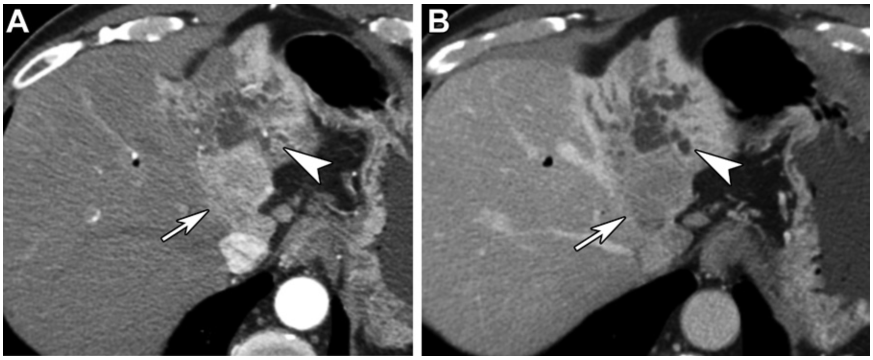


Figure 17. Intraductal-growing–type iCCA in a 55-year-old man with obstructive jaundice. Axial contrast-enhanced CT images of the liver in the arterial **(A)** and portal venous **(B)** phases show a focal mass (arrow) in the left hepatic lobe, with continuous extension into the left hepatic duct and homogeneous enhancement in the arterial phase with washout in the portal venous phase. There is also associated intrahepatic biliary dilatation (arrowhead). Pathologic analysis results confirmed intraductal-growing–type iCCA.

A recent study showed that contrast-enhanced CT texture analysis might help predict the *IDH1* mutation status of iCCAs (65). iCCAs with *KRAS* mutations demonstrate high metabolic tumor volume at ^{18}F–fluorodeoxyglucose (FDG) PET owing to increased glucose transporter protein 1 (GLUT1) expression and carry a worse prognosis (66). Qualitative imaging features such as necrosis, satellite nodules, and vascular encasement are associated with increased mortality (67).

The degree of enhancement at delayed phase CT correlated with fibrous stroma and increased frequency of perineural invasion (68). Hypovascular MF iCCAs tended to be more aggressive tumors with a poorer prognosis than hypervascular tumors (Fig 16) (69,70). MF iCCAs showing restricted diffusion in less than one-third of the tumor volume were associated with advanced TNM staging, frequent metastases, and low overall survival rates (71).

More recently, radiomics has been used to predict prognosis, lymph node metastasis, and response to radioembolization (72).

Management of iCCAs

Molecular profiling of iCCAs is being increasingly used for accurate diagnosis of iCCAs to facilitate optimal management (3,56). Complete surgical resection is considered curative for iCCAs. However, more than 50% of patients who undergo surgery experience recurrence within 12 months. Neoadjuvant chemotherapy, local-regional control with transarterial chemoembolization and radioembolization with yttrium 90, and liver transplantation are other treatment options for iCCAs. These tumors have a dismal 5-year survival rate of 7%–20% (3).

Targeted therapies based on mutational status in various iCCA subtypes are on the horizon and include inhibitors of *IDH1* (AG-120 [ivosidenib]), *IDH2* (AG-221), and both (AG-881) (56). Drugs targeting *FGFR2* genetic aberrations, including pemigatinib and infigratinib, have been recently approved to treat *FGFR2*-positive locally advanced or metastatic iCCAs. Clinical trials involving immune checkpoint inhibitors such as anti–programmed cell death protein 1 (PD-1), anti–programmed cell death ligand 1 (PD-L1), and anti–cytotoxic T-lymphocyte antigen-4 (CTLA-4) are ongoing (3,56).

Large-duct iCCAs are more likely to be associated with higher carbohydrate antigen 19-9 (CA 19-9) level, increased prevalence of lymphadenopathy, and higher stage at diagnosis than small-duct iCCAs and often have a worse prognosis than small-duct iCCAs (73). Small-duct iCCAs with *IDH1*/*IDH2* mutations are associated with a favorable prognosis, with improved survival compared with those without these mutations.

Conclusion

Primary hepatic neoplasms can be categorized into specific subtypes on the basis of characteristic histopathologic features, genetic abnormalities, and tumor pathways, which determine biologic behavior, response to therapy, and prognosis. Specific MRI characteristics of HCAs allow imaging-based distinction of steatotic and nonsteatotic HCAs. The variegated pathologic features of HCCs are reflected in the heterogeneity of cross-sectional imaging findings. While clear cell and steatohepatitic-type HCCs frequently show steatosis at MRI, MTM subtype HCCs are characterized by multifocal tumors and venous tumor thrombosis. MF iCCAs manifest as hypovascular tumors with or without bile duct involvement and show delayed enhancement owing to abundant fibrous stroma, portending a poor prognosis.

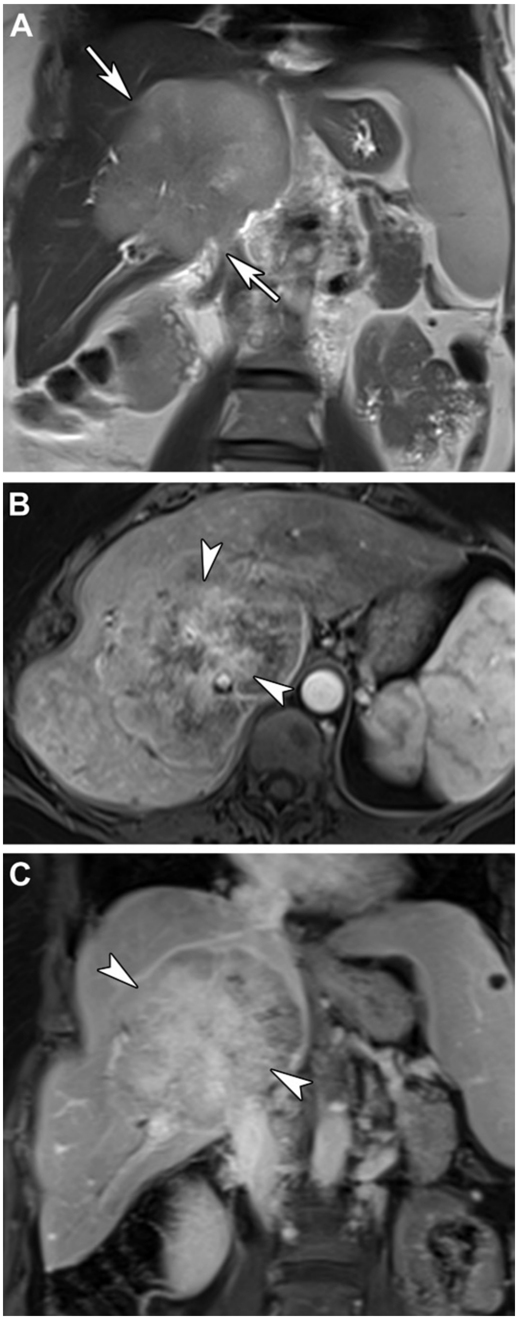


Figure 18. *IDH1* (isocitrate dehydrogenase)–mutated parenchymal-type MF iCCA in an 81-year-old woman. **(A)** Coronal T2-weighted MR image of the liver shows a large T2-hyperintense liver mass (arrows) with mild intrahepatic biliary dilatation in the right lobe. **(B)** Axial contrast-enhanced arterial phase fat-saturated T1-weighted MR image shows heterogeneous arterial enhancement of the mass (arrowheads), predominantly in the central aspect. **(C)** Coronal contrast-enhanced portal venous phase fat-saturated T1-weighted MR image shows progressive enhancement involving the entire tumor (arrowheads). The mass was proved to be *IDH1*-mutated iCCA at pathologic analysis.

A better understanding of tumor pathology, genetics, and pathways helps radiologists play an important role in multidisciplinary management of patients with HCAs, HCCs, and iCCAs.

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